

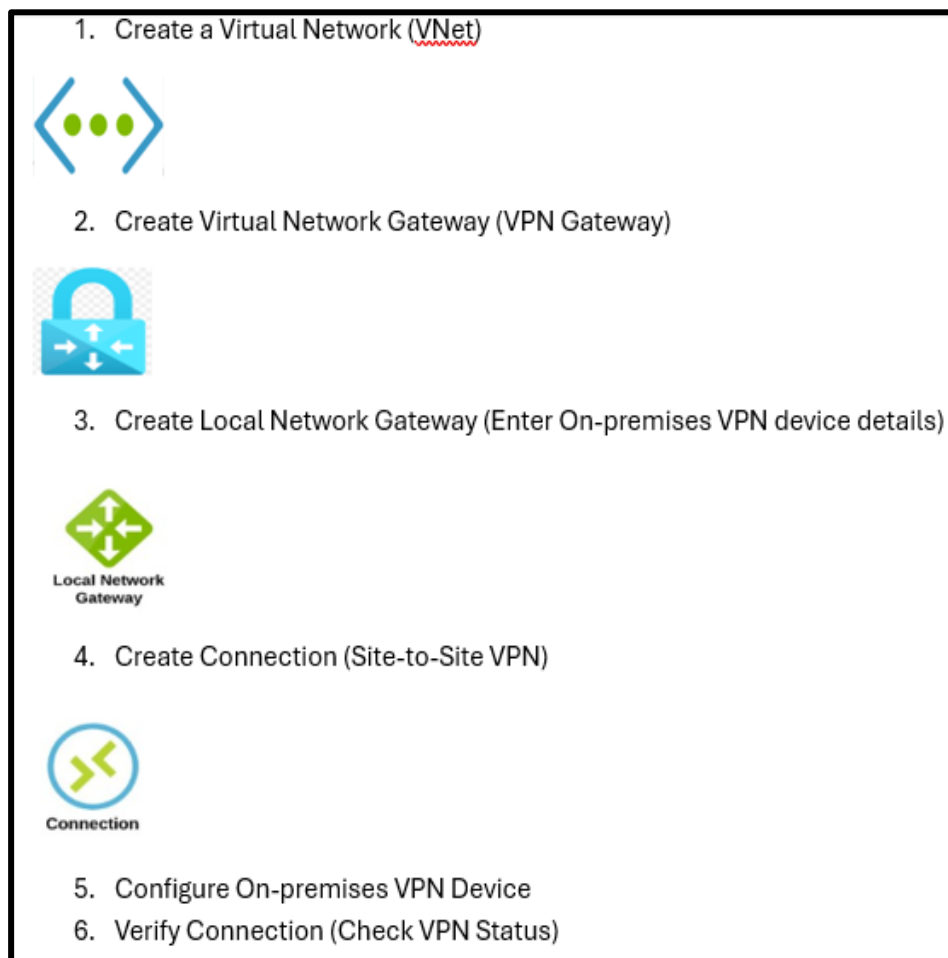
## To create a **Site-to-Site VPN** in **Microsoft Azure**

Need to establish a secure connection between your on-premises network (or another cloud environment) and Azure virtual network (VNet). Here's a step-by-step guide to set this up:

### Prerequisites

1. **Azure Subscription:** Need an Azure account.
2. **On-Premises VPN Device:** Need a compatible on-premises VPN device or gateway (like a Cisco ASA, Fortinet, or any supported VPN appliance).
3. **Public IP Address:** On-premises gateway should have a public IP address.
4. **Virtual Network (VNet):** Need an Azure VNet to connect to.

### Flowchart Diagram Example



### Steps to Create a Site-to-Site VPN in Azure

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## Step 1: Create a Virtual Network (VNet) in Azure

1. **Log into Azure Portal** ([portal.azure.com](https://portal.azure.com)).
2. In the left-hand menu, click **Create a resource**.
3. Search for **Virtual Network** and select it.
4. Click **Create**.
5. In the **Create Virtual Network** pane:
  - **Subscription**: Choose subscription.
  - **Resource group**: Choose an existing resource group or create a new one.
  - **Name**: Provide a name for the VNet (e.g., AzureVNet).
  - **Region**: Choose a region close to on-premises network.

The screenshot shows the 'Create virtual network' page in the Azure Portal. The breadcrumb navigation at the top reads 'Home > Virtual networks >'. The page title is 'Create virtual network' followed by an ellipsis. Below the title are tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. A descriptive paragraph explains that a Virtual Network (VNet) is the fundamental building block for a private network in Azure, enabling communication between Azure resources, the internet, and on-premises networks. Below this is a 'Project details' section with instructions to select a subscription and resource group. The 'Subscription' dropdown is set to 'Azure subscription 1' and the 'Resource group' dropdown is set to '(New) RG'. Below the resource group dropdown is a 'Create new' link. The 'Instance details' section contains the 'Virtual network name' field set to 'VNET' and the 'Region' dropdown set to '(Asia Pacific) Central India'. Below the region dropdown is a 'Deploy to an Azure Extended Zone' link. At the bottom of the form are three buttons: 'Previous', 'Next', and 'Review + create'.

Home > Virtual networks >

### Create virtual network ...

Basics Security IP addresses Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more.](#)

#### Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \* Azure subscription 1

Resource group \* (New) RG

[Create new](#)

#### Instance details

Virtual network name \* VNET

Region \* (Asia Pacific) Central India

[Deploy to an Azure Extended Zone](#)

Previous Next Review + create

- **IP Addressing**: Define the address space (e.g., 10.0.0.0/16).

- **Subnets:** Create a subnet (e.g., 10.0.1.0/24).

Home > Virtual networks >

## Create virtual network

Basics Security IP addresses Tags Review + create

Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. [Learn more](#)

Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. [Learn more](#)

+ Add a subnet

10.0.0.0/16 [Delete address space](#)

10.0.0.0 /16 65,536 addresses

10.0.0.0 - 10.0.255.255

| Subnets | IP address range      | Size                | NAT gateway |
|---------|-----------------------|---------------------|-------------|
| VNETSUB | 10.0.1.0 - 10.0.1.255 | /24 (256 addresses) | -           |

Add IPv4 address space

Previous Next Review + create

6. After reviewing the configuration, click **Create**.

Home > Virtual networks >

## Create virtual network

Basics Security IP addresses Tags Review + create

[View automation template](#)

### Basics

|                |                      |
|----------------|----------------------|
| Subscription   | Azure subscription 1 |
| Resource Group | RG                   |
| Name           | VNET                 |
| Region         | Central India        |

### Security

|                               |          |
|-------------------------------|----------|
| Azure Bastion                 | Disabled |
| Azure Firewall                | Disabled |
| Azure DDoS Network Protection | Disabled |

### IP addresses

|               |                                       |
|---------------|---------------------------------------|
| Address space | 10.0.0.0/16 (65,536 addresses)        |
| Subnet        | VNETSUB (10.0.1.0/24) (256 addresses) |

### Tags

Previous Next Create

## Step 2: Create a Virtual Network Gateway

1. In the Azure portal, search for **Virtual network gateways**.
2. Click **Add** to create a new gateway.
3. In the **Create virtual network gateway** pane:
  - **Subscription**: Select the subscription where VNet is located.
  - **Resource Group**: Select the same resource group as VNet.
  - **Region**: Select the same region as VNet.
  - **Name**: Provide a name for the gateway (e.g., AzureVNetGateway).
  - **Gateway Type**: Select **VPN**.
  - **VPN Type**: Choose **Route-based** (recommended for most scenarios).
  - **SKU**: Select a SKU (e.g., **VpnGw1** or **VpnGw2** depending on needs).
  - **Virtual Network**: Select the VNet created earlier.

Home > Virtual network gateways >

### Create virtual network gateway

Basics Tags Review + create

Azure has provided a planning and design guide to help you configure the various VPN gateway options. [Learn more](#)

**Project details**

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \* Azure subscription 1

Resource group ⓘ RG (derived from virtual network's resource group)

**Instance details**

Name \* AzureVNetGateway ✓

Region \* Central India  
[Deploy to an Azure Extended Zone](#)

Gateway type \* ⓘ ☒ VPN ☐ ExpressRoute

SKU \* ⓘ VpnGw1

Generation ⓘ Generation1

Virtual network \* ⓘ (New) VNET  
[Create virtual network](#)

ⓘ Only virtual networks in the currently selected subscription and region are listed.

Gateway subnet address range \* ⓘ 10.0.1.0/24 ✓  
10.0.1.0 - 10.0.1.255 (256 addresses)

- **Public IP Address**: Select **Create new** and give it a name (e.g., AzureVPNPublicIP).
- **Configure BGP**: Choose **Disabled** unless you are using BGP for routing.

[Home](#) > [Virtual network gateways](#) >

## Create virtual network gateway

Subscription: [vpngw1](#)

Generation: [Generation1](#)

Virtual network: [\(New\) VNET](#)  
[Create virtual network](#)

**Only virtual networks in the currently selected subscription and region are listed.**

Gateway subnet address range: [10.0.1.0/24](#)  
 10.0.1.0 - 10.0.1.255 (256 addresses)

Public IP address

Public IP address: ☒ Create new ☐ Use existing

Public IP address name: [AzureVPNPublicIP](#)

Public IP address SKU: Standard

Assignment: ☐ Dynamic ☒ Static

Enable active-active mode: ☐ Enabled ☒ Disabled

Configure BGP: ☐ Enabled ☒ Disabled

Authentication Information (Preview)

Enable Key Vault Access: ☐ Enabled ☒ Disabled

Azure recommends using a validated VPN device with your virtual network gateway. To view a list of validated devices and instructions for configuration, refer to Azure's [documentation](#) regarding validated VPN devices.

[Review + create](#) [Previous](#) [Next : Tags >](#) [Download a template for automation](#)

4. Click **Review + Create**, and after validation, click **Create**.

[Home](#) > [Virtual network gateways](#) >

## Create virtual network gateway

[Validation passed](#)

[Basics](#) [Tags](#) [Review + create](#)

**Basics**

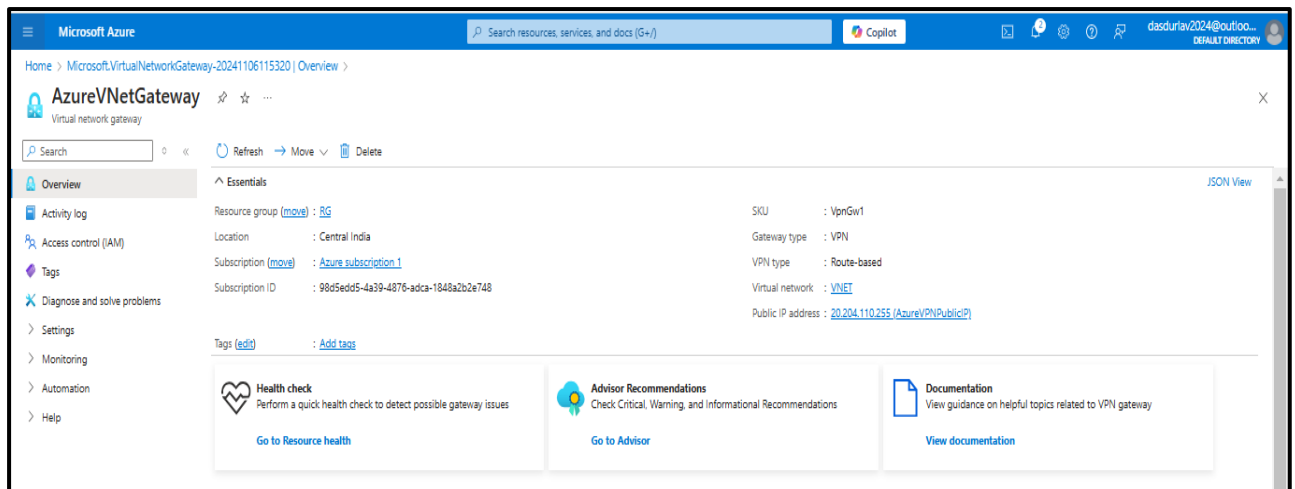
|                           |                             |
|---------------------------|-----------------------------|
| Subscription              | Azure subscription 1        |
| Resource group            | RG                          |
| Name                      | AzureVNetGateway            |
| Region                    | Central India               |
| SKU                       | VpnGw1                      |
| Generation                | Generation1                 |
| Virtual network           | VNET                        |
| Subnet                    | GatewaySubnet (10.0.1.0/24) |
| Gateway type              | Vpn                         |
| VPN type                  | RouteBased                  |
| Enable active-active mode | Disabled                    |
| Configure BGP             | Disabled                    |
| Public IP address         | AzureVPNPublicIP            |

**Tags**

None

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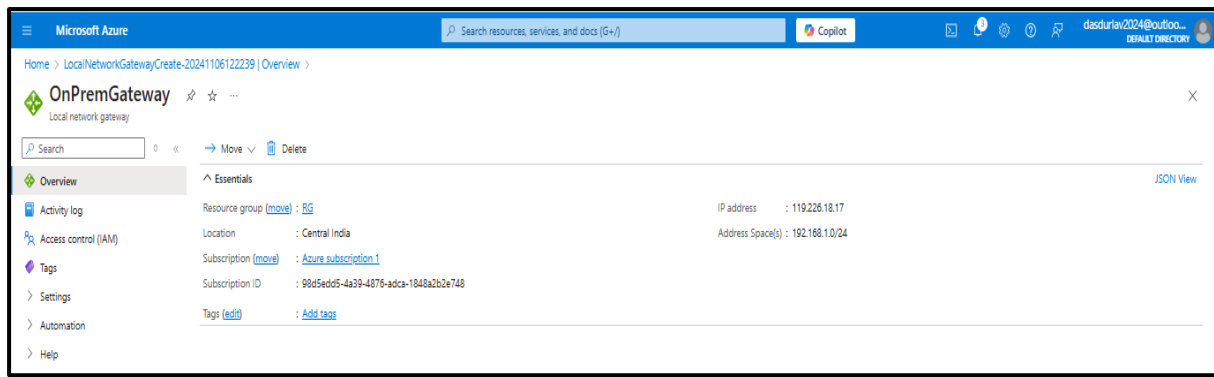
**Note:** The Virtual Network Gateway takes 30 minutes or more to deploy, depending on the SKU.



### Step 3: Configure the Local Network Gateway (On-Premises VPN Device)

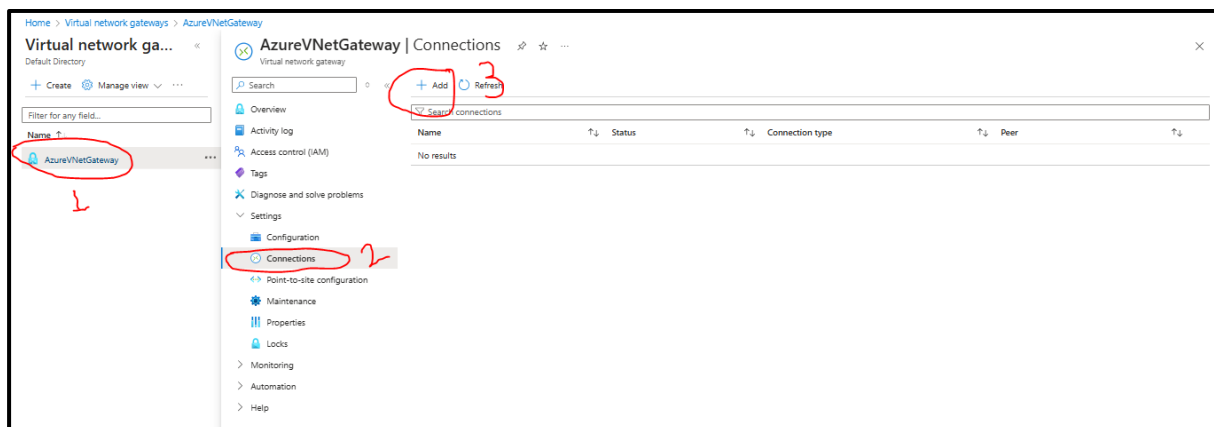
1. In the Azure portal, search for **Local network gateways**.
2. Click **Add** to create a new local network gateway.
3. In the **Create local network gateway** pane:
  - **Subscription:** Select the subscription.
  - **Resource Group:** Select the same resource group.
  - **Name:** Provide a name for the local network gateway (e.g., OnPremGateway).
  - **IP Address:** Enter the public IP address of on-premises VPN device.
  - **Address Space:** Provide the IP address range of on-premises network (e.g., 192.168.1.0/24).

4. Click **Review + Create** and then **Create**.



## Step 4: Create the VPN Connection

1. In the Azure portal, navigate to the **Virtual Network Gateway** you created earlier.
2. Under **Settings**, click **Connections**.
3. Click **Add** to create a new connection.



4. In the **Add connection** pane:
  - **Name:** Provide a name for the connection (e.g., SiteToSiteConnection).
  - **Connection Type:** Select **Site-to-Site (IPsec)**.
  - **Virtual Network Gateway:** This is pre-populated with your virtual network gateway.
  - **Region:** Select region

Home > Virtual network gateways > AzureVNetGateway | Connections >

## Create connection ...

Basics Settings Tags Review + create

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute.  
[Learn more about VPN Gateway](#) [Learn more about ExpressRoute](#)

**Project details**

Subscription \* Azure subscription 1

Resource group \* RG  
[Create new](#)

**Instance details**

Connection type \* Site-to-site (IPsec)

Name \* SiteToSiteConnection ✓

Region \* Central India

[Review + create](#) [Previous](#) [Next : Settings >](#) [Download a template for automation](#)

- **Local Network Gateway:** Select the local network gateway (you're on-premises device) you created earlier.
- **Shared Key:** Enter a shared secret (pre-shared key or PSK) that is also configured on your on-premises VPN device.
- **IKE Protocol:** Select IKE version (IKEv1/IKEv2)
- **IPsec/IKE policy:** Select policy mode (Default/Custom)
- **Policy based Traffic:** Enable/Disable the policy-based traffic selector.
- **Connection Mode:** Select connection mode (Default/Initiator only/Responder only)



Home > Virtual network gateways > AzureVNetGateway | Connections >

## Create connection

Basics Settings Tags Review + create

**Virtual network gateway**

To use a virtual network with a connection, it must be associated to a virtual network gateway.

Virtual network gateway \* ⓘ AzureVNetGateway

Local network gateway \* ⓘ OnPremGateway

Authentication Method ⓘ ☒ Shared Key(PSK) ☐ Key Vault Certificate (Preview)

Shared Key(PSK) \* ⓘ  ✓ ⓘ

IKE Protocol ⓘ ☐ IKEv1 ☒ IKEv2

Use Azure Private IP Address ⓘ ☐

Enable BGP ⓘ ☐

IPsec / IKE policy ⓘ Default Custom

Use policy based traffic selector ⓘ Enable Disable

DPD timeout in seconds \* ⓘ

Connection Mode ⓘ ☒ Default ☐ InitiatorOnly ☐ ResponderOnly

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- Click **Next**.

Home > Virtual network gateways > AzureVNetGateway | Connections >

## Create connection

✓ Validation passed

Basics Settings Tags Review + create

**Basics**

|                 |                      |
|-----------------|----------------------|
| Subscription    | Azure subscription 1 |
| Resource Group  | RG                   |
| Region          | Central India        |
| Connection type | Site-to-site (IPsec) |
| Connection name | SiteToSiteConnection |

**Settings**

|                                   |                  |
|-----------------------------------|------------------|
| Virtual network gateway           | AzureVNetGateway |
| Local network gateway             | OnPremGateway    |
| IKE Protocol                      | IKEv2            |
| IPsec / IKE policy                | Default          |
| Use policy based traffic selector | Disable          |
| DPD timeout in seconds            | 45               |
| Connection Mode                   | Default          |
| Shared Key(PSK)                   | VPNT-shooting    |

**Tags**

None

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- Click **Review + Create** and then **Create**

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## Step 5: Configure Your On-Premises VPN Device

Configure on-premises VPN device to establish the IPsec tunnel with Azure. The specific configuration depends on the device you're using, but here's an overview of the typical settings:

- **VPN Type:** IPsec/IKEv2
- **Gateway IP:** Public IP of the Azure VPN gateway.
- **Shared Key:** The pre-shared key (PSK) you specified in the Azure VPN connection.
- **Local Network:** on-premises subnet (e.g., 192.168.1.0/24).
- **Remote Network:** The Azure VNet's subnet (e.g., 10.0.1.0/24).
- **IKE Version:** Version 2 (IKEv2).
- **IPsec Encryption:** Ensure the encryption settings match what is supported by Azure (e.g., AES256).

For Example, FortiGate Firewall showcase here

The screenshot shows the 'New VPN Tunnel' configuration window in FortiGate. The 'Name' field is set to 'Test' and the 'Comments' field is empty. The 'Network' section is expanded, showing the following settings: IP Version is set to 'IPv4'; Remote Gateway is set to 'Static IP Address' with an IP Address of '119.226.18.17'; Interface is set to 'Airtel 50 mbps (wan1)'; Local Gateway is disabled; Mode Config is disabled; NAT Traversal is set to 'Enable'; Keepalive Frequency is set to '10'; Dead Peer Detection is set to 'On Demand'; DPD retry count is set to '3'; DPD retry interval is set to '20' seconds; Forward Error Correction is disabled for both Egress and Ingress. The 'Authentication' section is also expanded, showing the Method set to 'Pre-shared Key' and the Pre-shared Key field. The IKE Version is set to '2'. The 'OK' and 'Cancel' buttons are at the bottom.

**Phase 1 configuration**

New VPN Tunnel

Phase 1 Proposal ➕ Add

|            |            |                |           |   |
|------------|------------|----------------|-----------|---|
| Encryption | AES128     | Authentication | SHA256    | ✕ |
| Encryption | AES256     | Authentication | SHA256    | ✕ |
| Encryption | AES128GCM  | PRF ⓘ          | PRFSHA256 | ✕ |
| Encryption | AES256GCM  | PRF ⓘ          | PRFSHA384 | ✕ |
| Encryption | CHACHA20Pv | PRF ⓘ          | PRFSHA256 | ✕ |

Diffie-Hellman Groups

|                             |                                        |                                       |                             |                             |                             |
|-----------------------------|----------------------------------------|---------------------------------------|-----------------------------|-----------------------------|-----------------------------|
| <input type="checkbox"/> 32 | <input type="checkbox"/> 31            | <input type="checkbox"/> 30           | <input type="checkbox"/> 29 | <input type="checkbox"/> 28 | <input type="checkbox"/> 27 |
| <input type="checkbox"/> 21 | <input type="checkbox"/> 20            | <input type="checkbox"/> 19           | <input type="checkbox"/> 18 | <input type="checkbox"/> 17 | <input type="checkbox"/> 16 |
| <input type="checkbox"/> 15 | <input checked="" type="checkbox"/> 14 | <input checked="" type="checkbox"/> 5 | <input type="checkbox"/> 2  | <input type="checkbox"/> 1  |                             |

Key Lifetime (seconds)

86400

Local ID

## Phase 2 configuration

New VPN Tunnel

New Phase 2 ✓ ↺

Name

Test

Comments

Comments

Local Address

addr\_subnet 0.0.0.0/0.0.0.0

Remote Address

addr\_subnet 0.0.0.0/0.0.0.0

Advanced...

Phase 2 Proposal ➕ Add

|            |        |                |        |   |
|------------|--------|----------------|--------|---|
| Encryption | AES128 | Authentication | SHA1   | ✕ |
| Encryption | AES256 | Authentication | SHA1   | ✕ |
| Encryption | AES128 | Authentication | SHA256 | ✕ |
| Encryption | AES256 | Authentication | SHA256 | ✕ |

Enable Replay Detection ☒

Enable Perfect Forward Secrecy (PFS) ☒

Diffie-Hellman Group

|                             |                                        |                                       |                             |                             |                             |
|-----------------------------|----------------------------------------|---------------------------------------|-----------------------------|-----------------------------|-----------------------------|
| <input type="checkbox"/> 32 | <input type="checkbox"/> 31            | <input type="checkbox"/> 30           | <input type="checkbox"/> 29 | <input type="checkbox"/> 28 | <input type="checkbox"/> 27 |
| <input type="checkbox"/> 21 | <input type="checkbox"/> 20            | <input type="checkbox"/> 19           | <input type="checkbox"/> 18 | <input type="checkbox"/> 17 | <input type="checkbox"/> 16 |
| <input type="checkbox"/> 15 | <input checked="" type="checkbox"/> 14 | <input checked="" type="checkbox"/> 5 | <input type="checkbox"/> 2  | <input type="checkbox"/> 1  |                             |

Local Port

All ☒

Remote Port

All ☒

Protocol

All ☒

Auto-negotiate

☐

Autokey Keep Alive

☐

Key Lifetime

Seconds

43200

OK Cancel

**Note: Select Phase 1 and Phase 2 configuration as per Azure end. If not then connection not stablished.**

Check your VPN device's documentation for detailed setup steps.

---

## Step 6: Test the VPN Connection

Once everything is set up, check the connection status in the **Azure portal** under the **Connection** tab in the Virtual Network Gateway.

- **Status:** The connection should show as **Connected**.
- To verify connectivity, try to ping a resource in Azure (e.g., a virtual machine in the VNet) from your on-premises network.

## Troubleshooting

- Ensure that the shared key (PSK) matches on both sides.
  - Double-check the IP address ranges for both Azure and on-premises networks to avoid overlap.
  - Check firewall rules to ensure that UDP ports 500 and 4500 are open for IKE/IPsec negotiation.
  - If using BGP, make sure BGP settings are configured correctly on both sides.
- 

## Optional Steps: Configure Route Tables (if needed)

1. **On-Premises Routing:** Ensure that on-premises network has a route to reach the Azure VNet via the VPN tunnel.
2. **Azure Route Tables:** Need to configure route tables in Azure to properly route traffic from VNet to the on-premises network.